

Deep Generative Models: Variational Auto Encoders

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René Vidal

Director of the Center for Innovation in Data Engineering and Science (IDEAS)

Rachleff University Professor, University of Pennsylvania

Amazon Scholar & Chief Scientist at NORCE



The story up till now

- **Step 1:** We set out with our original goal of learning a model p_θ that gives maximum likelihood to our datapoints $\mathbf{x}_i$.
- **Step 2:** We introduced latent variables $\mathbf{z}$ such that $\mathbf{z} \sim p(\mathbf{z})$ and $\mathbf{x} | \mathbf{z} \sim p(\mathbf{x} | \mathbf{z})$, which gave us the marginalization $p(\mathbf{x}) = \int p(\mathbf{x} | \mathbf{z})p(\mathbf{z}) d\mathbf{z}$
 - Step 2a: When we assumed $p(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$, and $p(\mathbf{x} | \mathbf{z}) = \mathcal{N}(\mathbf{x}; \mathbf{W}\mathbf{z} + \mathbf{b}, \sigma^2 \mathbf{I})$, we could solve for $(\mathbf{W}, \mathbf{b}, \sigma^2)$ in closed form! This gave us PPCA.
- **Step 3:** We set up variational inference because, sadly, not everything in life is Gaussian and linear. This gave us a new Evidence Lower Bound (ELBO) objective:

$$\max_{\theta} \sum_{i=1}^N \log p_{\theta}(\mathbf{x}_i) = \max_{\theta} \max_q \sum_{i=1}^N \int q(\mathbf{z} | \mathbf{x}_i) \log \frac{p_{\theta}(\mathbf{x}_i, \mathbf{z})}{q(\mathbf{z} | \mathbf{x}_i)} d\mathbf{z}$$

- Step 3a: When the integral is easy to evaluate, we can alternate between optimizing w.r.t. θ with q fixed and vice versa, leading to the Expectation Maximization (EM) algorithm.

The story continues with Variational Autoencoders

- Before introducing VAEs formally, let us decompose the ELBO further

$$\begin{aligned}\max_{\theta} \sum_i \log p_{\theta}(\mathbf{x}_i) &= \max_{\theta} \max_q \sum_{i=1}^N \int q(\mathbf{z}|\mathbf{x}_i) \log \frac{p_{\theta}(\mathbf{x}_i, \mathbf{z})}{q(\mathbf{z}|\mathbf{x}_i)} d\mathbf{z} \\ &\geq \max_{\theta, \psi} \sum_i \mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z}|\mathbf{x}_i)} \log \frac{p_{\theta}(\mathbf{x}_i, \mathbf{z})}{q_{\psi}(\mathbf{z}|\mathbf{x}_i)} \quad \text{Evidence Lower Bound (ELBO)} \\ &= \max_{\theta, \psi} \sum_i \mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z}|\mathbf{x}_i)} \log p_{\theta}(\mathbf{x}_i|\mathbf{z}) \frac{p(\mathbf{z})}{q_{\psi}(\mathbf{z}|\mathbf{x}_i)}\end{aligned}$$

$$= \max_{\theta, \psi} \sum_i \mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z}|\mathbf{x}_i)} \log p_{\theta}(\mathbf{x}_i|\mathbf{z}) + \mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z}|\mathbf{x}_i)} \log \frac{p(\mathbf{z})}{q_{\psi}(\mathbf{z}|\mathbf{x}_i)}$$

Given $\mathbf{x}$, sample latent $\mathbf{z}$ from q .

Given latent $\mathbf{z}$, sample $\mathbf{x}$ from p .

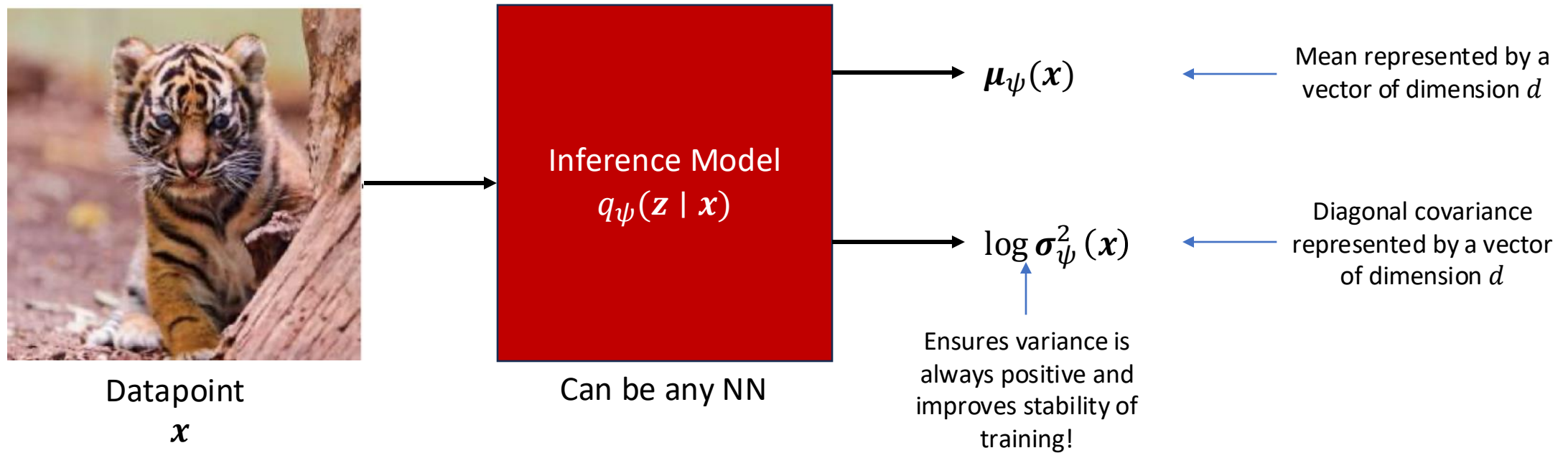
$$\begin{array}{c} \uparrow \\ -D_{KL}(q_{\psi}(\mathbf{z}|\mathbf{x})||p(\mathbf{z})) \end{array}$$

Measure average likelihood of $\mathbf{x}$ across all choices of $\mathbf{z}$

Variational AutoEncoders (VAEs): Setup

- We have three models we need to define for VAE model

1. **Inference model $q_{\psi}(\mathbf{z} | \mathbf{x})$** : We will define it as $q_{\psi}(\mathbf{z} | \mathbf{x}) = \mathcal{N}(\mathbf{z}; \boldsymbol{\mu}_{\psi}(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_{\psi}^2(\mathbf{x})))$, i.e., a normal distribution with learned mean and covariance



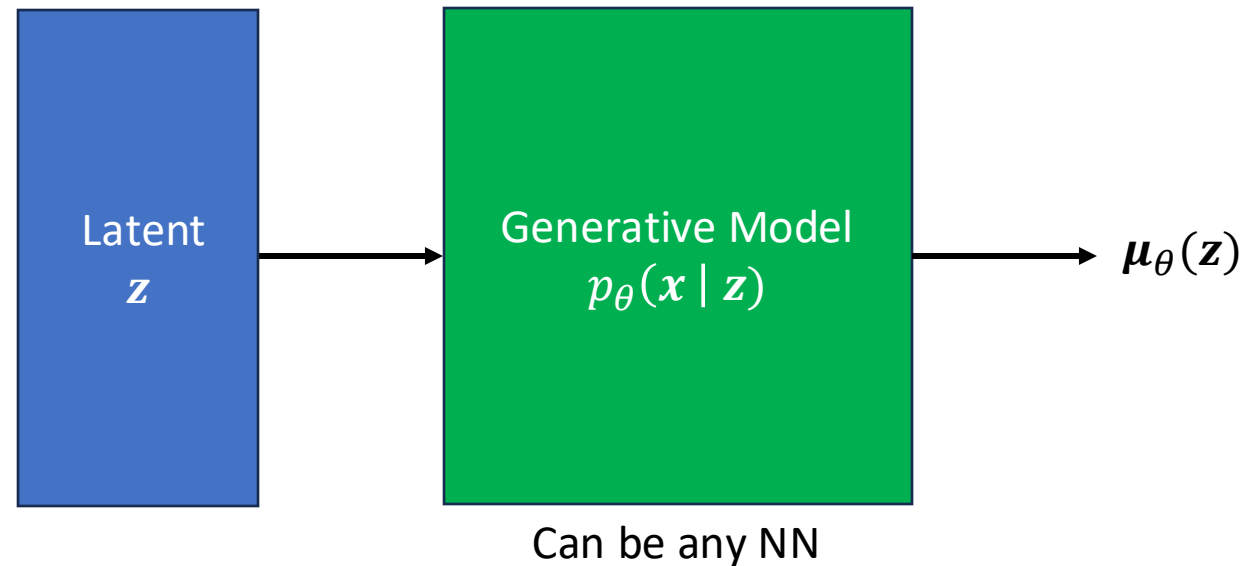
2. **Prior model $p(\mathbf{z})$** : We will define prior for latent variables as $p(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$

Variational AutoEncoders (VAEs): Setup

- We have three models we need to define for VAE model

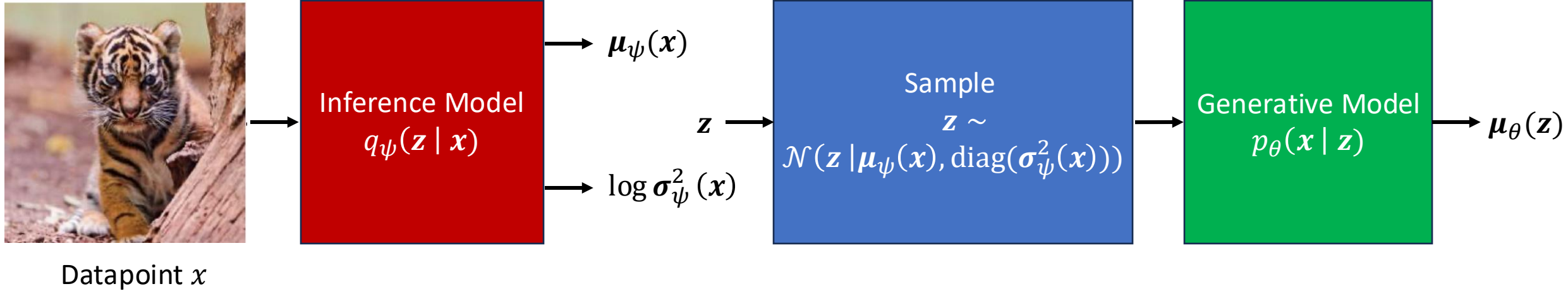
3. **Generative model $p_{\theta}(\mathbf{x} | \mathbf{z})$** : We will define it as

- $p_{\theta}(\mathbf{x} | \mathbf{z}) = \mathcal{N}(\mathbf{x}; \mu_{\theta}(\mathbf{z}), \eta^2 \mathbf{I})$, i.e., a normal distribution with learned mean and variance
- $p_{\theta}(\mathbf{x} | \mathbf{z}) = \text{Cat}(\mathbf{x}; \pi_{\theta}(\mathbf{z}))$, i.e., a categorical distribution with learned class probabilities



- Note this can be defined in many different ways, yielding different models (such as a categorical distribution over 255 values of each pixel)

Variational AutoEncoders: Training

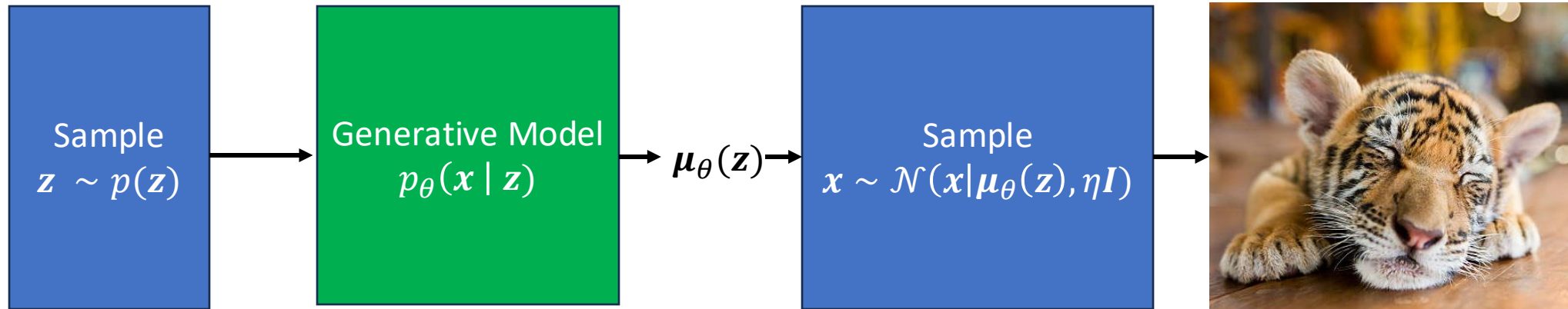


ELBO Objective

$$\mathbb{E}_{z \sim q_{\psi}(z|x)} [\log p_{\theta}(x | z)] - KL(q_{\psi}(z | x) || p(z))$$

Variational AutoEncoders: Sampling

- Suppose we have learned VAE using the ELBO loss (details to follow).
- Then, as a generative model, we just sample $\mathbf{z} \sim p(\mathbf{z})$ and use the fixed generative model $p_{\theta}(\mathbf{x} | \mathbf{z})$.



Computing the ELBO Loss

$$L_{\theta, \psi}(\mathbf{x}) := \underbrace{\mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z} | \mathbf{x})} \log p_{\theta}(\mathbf{x} | \mathbf{z})}_{\text{Term 1 (Reconstruction Error)}} + \underbrace{\mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z} | \mathbf{x})} \log \frac{p(\mathbf{z})}{q_{\psi}(\mathbf{z} | \mathbf{x})}}_{\text{Term 2 (KL Divergence)}}$$

- **Term 1 (Reconstruction Error)**

- Because $p_{\theta}(\mathbf{x} | \mathbf{z}) = \mathcal{N}(\mathbf{x} | \boldsymbol{\mu}_{\theta}(\mathbf{z}), \eta \mathbf{I})$, we have $\log p_{\theta}(\mathbf{x} | \mathbf{z}) = -\frac{1}{2\eta} \|\mathbf{x} - \boldsymbol{\mu}_{\theta}(\mathbf{z})\|_2^2 + \text{const.}$
- We can approximate the expectation over $\mathbf{z} \sim q_{\psi}(\mathbf{z} | \mathbf{x})$ by an average over $q_{\psi}(\mathbf{z} | \mathbf{x})$
- Recall that the expectation of a function $f(\mathbf{z})$ w.r.t. a random variable $\mathbf{z} \sim p$ can be approximated from i.i.d. samples $\mathbf{z}_j \sim p, j = 1, \dots, M$, via Monte Carlo averages

$$\mathbb{E}_{\mathbf{z} \sim p}[f(\mathbf{z})] = \int_{\mathbf{z}} f(\mathbf{z}) p(\mathbf{z}) d\mathbf{z} \approx \frac{1}{M} \sum_j f(\mathbf{z}_j)$$

- Applying the above formula to M i.i.d. samples $\mathbf{z}_j \sim q_{\psi}(\mathbf{z} | \mathbf{x})$ we get **reconstruction error**

$$\mathbb{E}_{\mathbf{z} \sim q_{\psi}}[\log p_{\theta}(\mathbf{x} | \mathbf{z})] \approx \frac{1}{M} \sum_j \log p_{\theta}(\mathbf{x} | \mathbf{z}_j) = -\frac{1}{2\eta M} \sum_j \|\mathbf{x} - \boldsymbol{\mu}_{\theta}(\mathbf{z}_j)\|_2^2 + \text{const.}$$

Computing the ELBO Loss

$$L_{\theta,\psi}(\mathbf{x}) := \underbrace{\mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z} | \mathbf{x})} \log p_{\theta}(\mathbf{x} | \mathbf{z})}_{\text{Term 1}} + \underbrace{\mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z} | \mathbf{x})} \log \frac{p(\mathbf{z})}{q_{\psi}(\mathbf{z} | \mathbf{x})}}_{\text{Term 2}}$$

- **Term 2 (Regularization to Prior)**

- The second term is a KL divergence between two Gaussians, which can be computed as

$$KL(\mathcal{N}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1) \parallel \mathcal{N}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)) = \frac{1}{2} \left(\log \left(\frac{\det \boldsymbol{\Sigma}_2}{\det \boldsymbol{\Sigma}_1} \right) - d + \text{trace}(\boldsymbol{\Sigma}_2^{-1} \boldsymbol{\Sigma}_1) + (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2)^{\top} \boldsymbol{\Sigma}_2^{-1} (\boldsymbol{\mu}_1 - \boldsymbol{\mu}_2) \right)$$

- Applying this formula to $q_{\psi}(\mathbf{z} | \mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}_{\psi}(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_{\psi}^2(\mathbf{x})))$ and $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ yields

$$KL(q_{\psi}(\mathbf{z} | \mathbf{x}) \parallel p(\mathbf{z})) = - \sum_{i=1}^d \log(\sigma_{i,\psi}(\mathbf{x})) + \frac{\|\boldsymbol{\sigma}_{\psi}(\mathbf{x})\|_2^2 + \|\boldsymbol{\mu}_{\psi}(\mathbf{x})\|_2^2}{2} + \text{constant}$$

- Note this term does not require any sampling w.r.t. $\mathbf{z}$ because the expectation is computed in closed form thanks for the KL formula for Gaussians.

Maximizing ELBO: How to Optimize?

- **ELBO objective:** We want to solve the following optimization problem:

$$\max_{\theta, \psi} \sum_i L_{\theta, \psi}(\mathbf{x}_i) = \max_{\theta, \psi} \sum_i \mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z} | \mathbf{x}_i)} \log(p_{\theta}(\mathbf{x}_i | \mathbf{z})) + KL(q_{\psi}(\mathbf{z} | \mathbf{x}) \parallel p(\mathbf{z}))$$

- $$= \max_{\theta, \psi} \sum_{i=1}^N -\frac{1}{2\eta M} \sum_{j=1}^M [\|\mathbf{x}_i - \boldsymbol{\mu}_{\theta}(\mathbf{z}_j)\|_2^2] - \sum_{i=1}^d \log(\sigma_{i, \psi}(\mathbf{x})) + \frac{\|\boldsymbol{\sigma}_{\psi}(\mathbf{x})\|_2^2 + \|\boldsymbol{\mu}_{\psi}(\mathbf{x})\|_2^2}{2} + \text{const}$$

- **Simple idea:** Just alternate gradient ascent w.r.t. θ, ψ on objective function

$$\theta_{k+1} = \theta_k + \alpha \frac{\delta L}{\delta \theta}(\theta_k, \psi_k)$$

$$\psi_{k+1} = \psi_k + \alpha \frac{\delta L}{\delta \psi}(\theta_k, \psi_k)$$

Stochastic Optimization of ELBO wrt θ

- **Issue:** Computing ∇_{θ} (ELBO) is not easy because of expectation w.r.t. latent z .
- **Solution:** Compute an unbiased estimator

$$\begin{aligned}\nabla_{\theta} L_{\theta, \psi}(\mathbf{x}) &= \nabla_{\theta} \mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z} | \mathbf{x})} [\log(p_{\theta}(\mathbf{x} | \mathbf{z}))] + \nabla_{\theta} \mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z} | \mathbf{x})} \left[\log \left(\frac{p(\mathbf{z})}{q_{\psi}(\mathbf{z} | \mathbf{x})} \right) \right] \\ &= \mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z} | \mathbf{x})} [\nabla_{\theta} \log p_{\theta}(\mathbf{x} | \mathbf{z})] \\ &= -\frac{1}{2\eta} \mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z} | \mathbf{x})} \nabla_{\theta} \|\mathbf{x} - \boldsymbol{\mu}_{\theta}(\mathbf{z})\|_2^2\end{aligned}$$

- We can take sample averages to compute an unbiased estimator
 - For each datapoint $\mathbf{x}$, compute $q_{\psi}(\mathbf{z} | \mathbf{x})$ through encoder, which gives $\boldsymbol{\mu}_{\psi}(\mathbf{x})$, $\boldsymbol{\sigma}_{\psi}^2(\mathbf{x})$
 - Sample $\mathbf{z}_j \sim \mathcal{N}(\boldsymbol{\mu}_{\psi}(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_{\psi}^2(\mathbf{x})))$, find $\boldsymbol{\mu}_{\theta}(\mathbf{z}_j)$ through decoder
 - Estimate gradient from M samples: $\nabla_{\theta} L_{\theta, \psi}(\mathbf{x}) \approx -\frac{1}{2\eta M} \sum_{j=1}^M \nabla_{\theta} \|\mathbf{x} - \boldsymbol{\mu}_{\theta}(\mathbf{z}_j)\|_2^2$

Stochastic Optimization of ELBO wrt ψ

- Here, we cannot just switch gradient and expectation because both are wrt to ψ

$$\nabla_{\psi} \mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z}|\mathbf{x})} [\log p_{\theta}(\mathbf{x}, \mathbf{z}) - \log q_{\psi}(\mathbf{z} | \mathbf{x})] \neq \mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z}|\mathbf{x})} \nabla_{\psi} [\log p_{\theta}(\mathbf{x}, \mathbf{z}) - \log q_{\psi}(\mathbf{z} | \mathbf{x})]$$

- **Reparameterization trick**

- Because $q_{\psi}(\mathbf{z} | \mathbf{x}) = N(\mathbf{z}; \boldsymbol{\mu}_{\psi}(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_{\psi}^2(\mathbf{x})))$, we can rewrite samples $\mathbf{z} \sim q_{\psi}(\mathbf{z} | \mathbf{x})$ as

$$\mathbf{z}_{\psi} = \boldsymbol{\mu}_{\psi}(\mathbf{x}) + \boldsymbol{\sigma}_{\psi}(\mathbf{x}) \odot \boldsymbol{\epsilon} \text{ for } \boldsymbol{\epsilon} \sim N(\mathbf{0}, \mathbf{I})$$

- With this change of variables, we can rewrite the gradient of the loss as:

$$\begin{aligned} \nabla_{\psi} L_{\theta, \psi}(\mathbf{x}) &= \nabla_{\psi} \mathbb{E}_{\mathbf{z} \sim q_{\psi}(\mathbf{z}|\mathbf{x})} [\log p_{\theta}(\mathbf{x}, \mathbf{z}) - \log q_{\psi}(\mathbf{z} | \mathbf{x})] \\ &= \nabla_{\psi} \mathbb{E}_{\boldsymbol{\epsilon} \sim N(\mathbf{0}, \mathbf{I})} [\log p_{\theta}(\mathbf{x}, \mathbf{z}_{\psi}) - \log q_{\psi}(\mathbf{z}_{\psi} | \mathbf{x})] \\ &= \mathbb{E}_{\boldsymbol{\epsilon} \sim N(\mathbf{0}, \mathbf{I})} \nabla_{\psi} [\log p_{\theta}(\mathbf{x}, \mathbf{z}_{\psi}) - \log q_{\psi}(\mathbf{z}_{\psi} | \mathbf{x})] \end{aligned}$$

Stochastic Optimization of ELBO wrt ψ

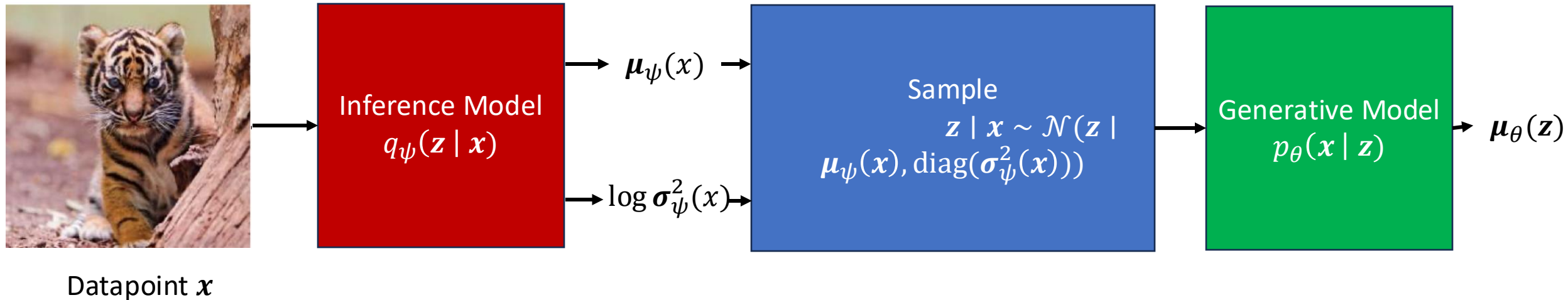
- Reparameterization trick

$$\nabla_{\psi} L_{\theta, \psi}(\mathbf{x}) = \mathbb{E}_{\epsilon \sim N(\mathbf{0}, I)} \nabla_{\psi} \left[-\frac{1}{2\eta} \|\mathbf{x} - \boldsymbol{\mu}_{\theta}(\mathbf{z}_{\psi})\|_2^2 \right] + \nabla_{\psi} \left[-\sum_{i=1}^d \log(\sigma_{i, \psi}(\mathbf{x})) + \frac{\|\boldsymbol{\sigma}_{\psi}(\mathbf{x})\|_2^2 + \|\boldsymbol{\mu}_{\psi}(\mathbf{x})\|_2^2}{2} \right]$$

- The computation of the gradient requires gradients of the encoder and decoder networks with respect to the network weights + chain rule.
- Once the gradient has been computed, we can compute the expectation via Monte Carlo sampling

Putting it all together

- Variational Autoencoder
 - We modelled inference and generative model as deep networks
 - We interpreted ELBO as an expected reconstruction error plus a KL-regularization to prior
 - Then, we rewrote the sampling in the latent space using the reparameterization trick
 - Finally, we derived stochastic gradient estimates to optimize the ELBO and learn a VAE



ELBO Objective

$$\mathbb{E}_{z \sim q_{\psi}(z|x)} [\log p_{\theta}(x | z)] - KL(q_{\psi}(z | x) || p(z))$$

VAEs in Action

- $q_{\psi}(\mathbf{z} \mid \mathbf{x}) = \mathcal{N}(\mathbf{z} \mid \boldsymbol{\mu}_{\psi}(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_{\psi}^2(\mathbf{x})))$
- $p(\mathbf{z}) = \mathcal{N}(\mathbf{z} \mid \mathbf{0}, \mathbf{I})$
- $p_{\theta}(\mathbf{x} \mid \mathbf{z}) = \text{Cat}(\mathbf{x} \mid \pi_{\theta}(\mathbf{z}))$

Note this is different from the model considered up till now!

- Encoder network: $\mathbf{x} \in \mathbb{R}^D \rightarrow \text{Linear}(D, 256) \rightarrow \text{LeakyReLU} \rightarrow \text{Linear}(256, 2d) \rightarrow$
split into $\mu \in \mathbb{R}^d, \log \sigma^2 \in \mathbb{R}^d$
- Decoder network: $\mathbf{z} \in \mathbb{R}^d \rightarrow \text{Linear}(d, 256) \rightarrow \text{LeakyReLU} \rightarrow \text{Linear}(256, D) \rightarrow$
softmax

VAE's for Generation of MNIST Digits

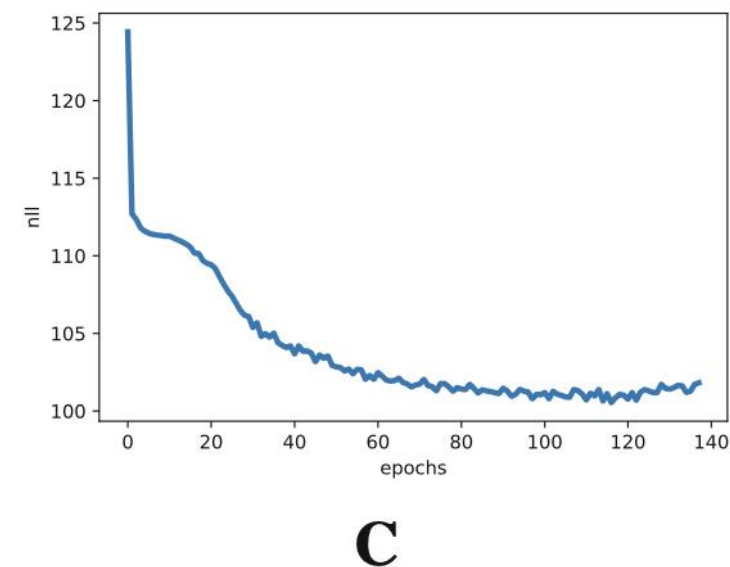


Fig. 4.4 An example of outcomes after the training: (a) Randomly selected real images. (b) Unconditional generations from the VAE. (c) The validation curve during training

VAEs: extensions

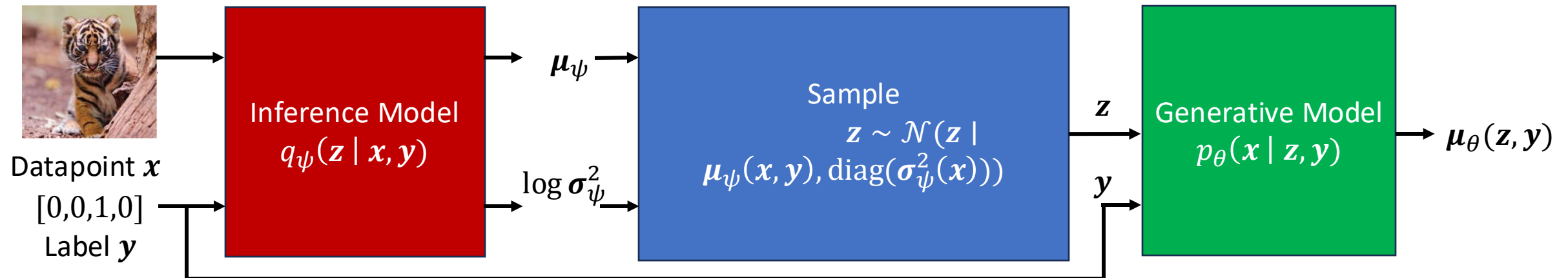
- So far, we have seen how to model $p(\mathbf{x})$ via VAEs.
- **Question:** What if we want conditional generation, i.e., sampling from $p(\mathbf{x}|\mathbf{y})$?
- **Answer:** Model $p(\mathbf{x}|\mathbf{y})$ via conditional VAE
- **Question:** What if we want both conditional generation $p(\mathbf{x}|\mathbf{y})$ and classification $p(\mathbf{y}|\mathbf{x})$?
- **Answer:** Model $p(\mathbf{x}, \mathbf{y})$ via joint VAE

Conditional VAE

- Regular VAE: start from $\log p_{\theta}(\mathbf{x})$ and derive its ELBO $\int q(\mathbf{z} | \mathbf{x}) \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z} | \mathbf{x})} d\mathbf{z}$
- Here, since we want to model $p_{\theta}(\mathbf{x} | \mathbf{y})$, let us derive ELBO for $\log p_{\theta}(\mathbf{x} | \mathbf{y})$

$$\max_{\theta} \log p_{\theta}(\mathbf{x} | \mathbf{y}) = \max_{\theta} \max_{\psi} \mathbb{E}_{q_{\psi}(\mathbf{z} | \mathbf{x}, \mathbf{y})} \log p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y}) - KL[q_{\psi}(\mathbf{z} | \mathbf{x}, \mathbf{y}) || p_{\theta}(\mathbf{z} | \mathbf{y})]$$

- Details in the next slide. Spoiler alert: the ELBO derivation is almost the same as before
- Architecture:



Conditional VAE: ELBO

- Let $q_\psi(\mathbf{z}|\mathbf{x}, \mathbf{y})$ be the variational distribution. Observe that

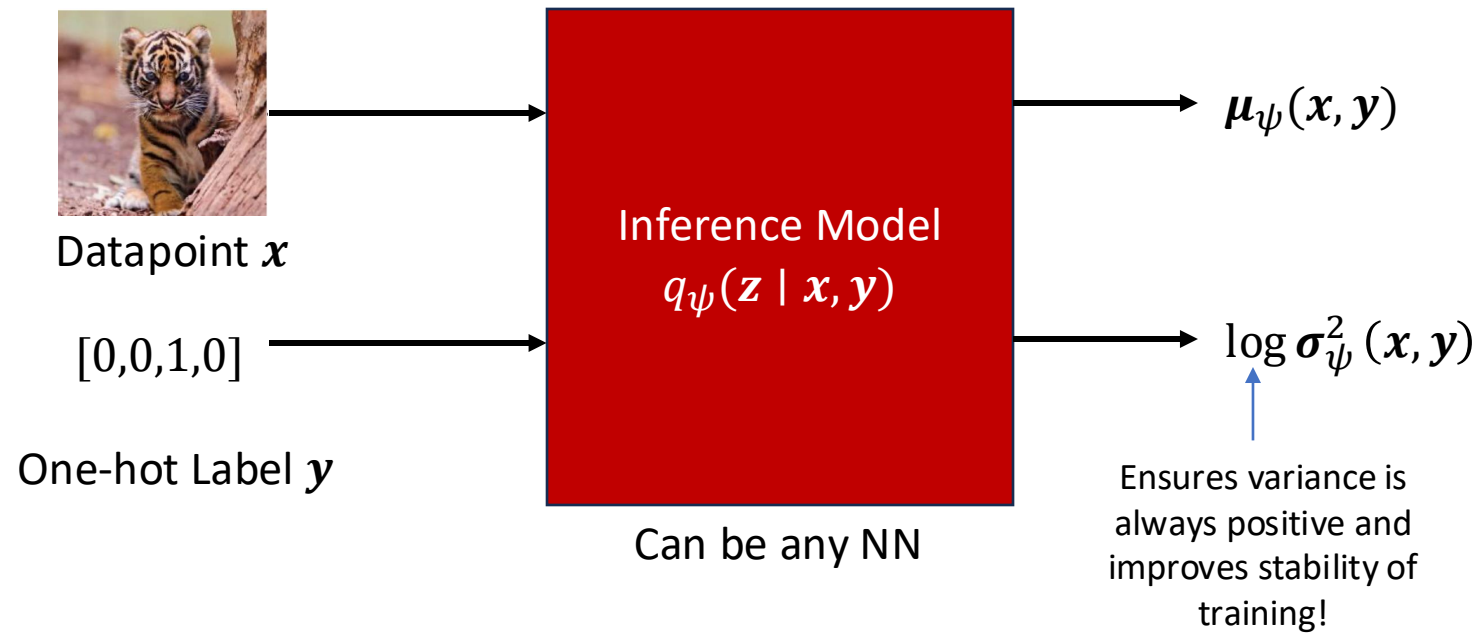
$$\begin{aligned}\log p_\theta(\mathbf{x} | \mathbf{y}) &= \int q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y}) \log p_\theta(\mathbf{x} | \mathbf{y}) d\mathbf{z} = \int q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y}) \log \frac{p_\theta(\mathbf{x}, \mathbf{z} | \mathbf{y})}{p_\theta(\mathbf{z} | \mathbf{x}, \mathbf{y})} d\mathbf{z} \\ &= \int q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y}) \log \frac{p_\theta(\mathbf{x}, \mathbf{z} | \mathbf{y})}{q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y})} \frac{q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y})}{p_\theta(\mathbf{z} | \mathbf{x}, \mathbf{y})} d\mathbf{z} \\ &= \boxed{\int q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y}) \log \frac{p_\theta(\mathbf{x}, \mathbf{z} | \mathbf{y})}{q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y})} d\mathbf{z}} + \boxed{\int q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y}) \log \frac{q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y})}{p_\theta(\mathbf{z} | \mathbf{x}, \mathbf{y})} d\mathbf{z}} \\ &\quad \text{Evidence Lower Bound (ELBO)} \quad \text{KL}[q(\mathbf{z} | \mathbf{x}, \mathbf{y}) || p_\theta(\mathbf{z} | \mathbf{x}, \mathbf{y})]\end{aligned}$$

$$\begin{aligned}\text{Therefore, } \max_{\theta} \log p_\theta(\mathbf{x} | \mathbf{y}) &\geq \max_{\theta} \max_{\psi} \mathbb{E}_{q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y})} \log \frac{p_\theta(\mathbf{x}, \mathbf{z} | \mathbf{y})}{q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y})} \\ &= \max_{\theta} \max_{\psi} \mathbb{E}_{q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y})} \log \frac{p_\theta(\mathbf{x} | \mathbf{z}, \mathbf{y}) p_\theta(\mathbf{z} | \mathbf{y})}{q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y})} \\ &= \max_{\theta} \max_{\psi} \mathbb{E}_{q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y})} \log p_\theta(\mathbf{x} | \mathbf{z}, \mathbf{y}) - \text{KL}[q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y}) || p_\theta(\mathbf{z} | \mathbf{y})]\end{aligned}$$

Conditional VAE: Setup

- We have three models we need to define for VAE model

1. **Inference model $q_\psi(\mathbf{z} \mid \mathbf{x}, \mathbf{y})$:** Defined as $q_\psi(\mathbf{z} \mid \mathbf{x}, \mathbf{y}) = \mathcal{N}(\boldsymbol{\mu}_\psi(\mathbf{x}, \mathbf{y}), \text{diag}(\boldsymbol{\sigma}_\psi^2(\mathbf{x}, \mathbf{y})))$, i.e., a normal distribution with learned mean and covariance



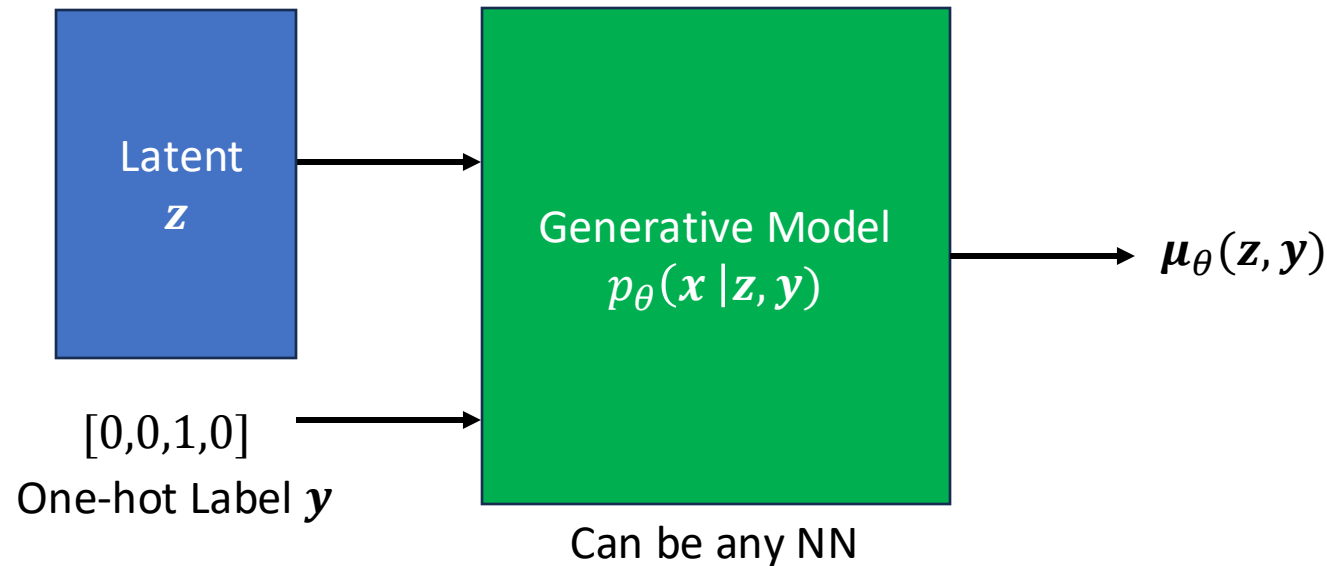
2. **Prior model $p(\mathbf{z} \mid \mathbf{y})$:** $p(\mathbf{z} \mid \mathbf{y}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$ or $p_\theta(\mathbf{z} \mid \mathbf{y}) = \mathcal{N}(\boldsymbol{\mu}_\theta(\mathbf{y}), \text{diag}(\boldsymbol{\sigma}_\theta^2(\mathbf{y})))$

Conditional VAE: Setup

- We have three models we need to define for VAE model

3. Generative model $p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y})$: Defined as

- $p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y}) = \mathcal{N}(\mu_{\theta}(\mathbf{z}, \mathbf{y}), \eta^2 \mathbf{I})$, i.e., a normal distribution with learned mean and variance
- $p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y}) = \text{Cat}(\pi_{\theta}(\mathbf{z}, \mathbf{y}))$, i.e., a categorical distribution with learned class probabilities



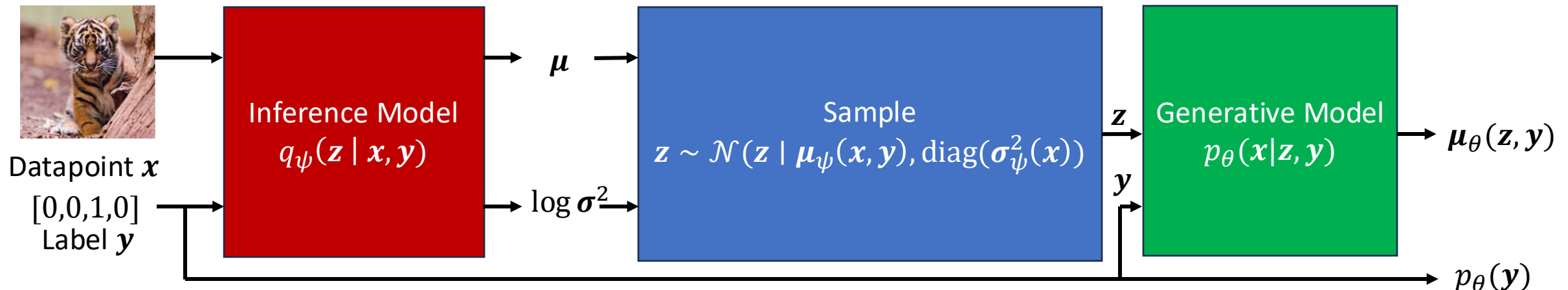
- Note this can be defined in many different ways, yielding different models (such as a categorical distribution over 255 values of each pixel)

Joint VAE

- Regular VAE: start from $\log p_{\theta}(\mathbf{x})$ and derive its ELBO $\int q(\mathbf{z} | \mathbf{x}) \log \frac{p_{\theta}(\mathbf{x}, \mathbf{z})}{q(\mathbf{z} | \mathbf{x})} d\mathbf{z}$
- Here, since we want to model $p_{\theta}(\mathbf{x}, \mathbf{y})$, let us derive ELBO for $\log p_{\theta}(\mathbf{x}, \mathbf{y})$

$$\begin{aligned} & \max_{\theta} \mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}} \log p_{\theta}(\mathbf{x}, \mathbf{y}) \\ &= \max_{\theta} \max_{\psi} \mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}} \left[\mathbb{E}_{q_{\psi}(\mathbf{z} | \mathbf{x}, \mathbf{y})} \log p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y}) - KL[q_{\psi}(\mathbf{z} | \mathbf{x}, \mathbf{y}) || p_{\theta}(\mathbf{z} | \mathbf{y})] + \log p_{\theta}(\mathbf{y}) \right] \end{aligned}$$

- Details in the next slide. Spoiler alert: the ELBO derivation is almost the same as before
- Architecture:



Joint VAE: ELBO

- Let $q_\psi(\mathbf{z}|\mathbf{x}, \mathbf{y})$ be the variational distribution. Observe that

$$\begin{aligned}\log p_\theta(\mathbf{x}, \mathbf{y}) &= \int q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y}) \log p_\theta(\mathbf{x}, \mathbf{y}) d\mathbf{z} = \int q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y}) \log \frac{p_\theta(\mathbf{x}, \mathbf{y}, \mathbf{z})}{p_\theta(\mathbf{z}|\mathbf{x}, \mathbf{y})} d\mathbf{z} \\ &= \int q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y}) \log \frac{p_\theta(\mathbf{x}, \mathbf{y}, \mathbf{z})}{q_\psi(\mathbf{z}|\mathbf{x}, \mathbf{y})} \frac{q_\psi(\mathbf{z}|\mathbf{x}, \mathbf{y})}{p_\theta(\mathbf{z}|\mathbf{x}, \mathbf{y})} d\mathbf{z} \\ &= \boxed{\int q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y}) \log \frac{p_\theta(\mathbf{x}, \mathbf{y}, \mathbf{z})}{q_\psi(\mathbf{z}|\mathbf{x}, \mathbf{y})} d\mathbf{z}} + \boxed{\int q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y}) \log \frac{q_\psi(\mathbf{z}|\mathbf{x}, \mathbf{y})}{p_\theta(\mathbf{z}|\mathbf{x}, \mathbf{y})} d\mathbf{z}} \\ &\quad \text{Evidence Lower Bound (ELBO)} \quad \text{KL}[q(\mathbf{z}|\mathbf{x}, \mathbf{y}) || p_\theta(\mathbf{z}|\mathbf{x}, \mathbf{y})]\end{aligned}$$

$$\begin{aligned}\text{Therefore, } \log p_\theta(\mathbf{x}, \mathbf{y}) &= \max_{\psi} \mathbb{E}_{q_\psi(\mathbf{z}|\mathbf{x}, \mathbf{y})} \log \frac{p_\theta(\mathbf{x}, \mathbf{y}, \mathbf{z})}{q_\psi(\mathbf{z}|\mathbf{x}, \mathbf{y})} \\ &= \max_{\psi} \mathbb{E}_{q_\psi(\mathbf{z}|\mathbf{x}, \mathbf{y})} \left[\log \frac{p_\theta(\mathbf{z}|\mathbf{y})}{q_\psi(\mathbf{z}|\mathbf{x}, \mathbf{y})} + \log p_\theta(\mathbf{x} | \mathbf{y}, \mathbf{z}) + \log p_\theta(\mathbf{y}) \right] \\ &= \max_{\psi} \mathbb{E}_{q_\psi(\mathbf{z}|\mathbf{x}, \mathbf{y})} \log p_\theta(\mathbf{x} | \mathbf{y}, \mathbf{z}) - \text{KL}[q_\psi(\mathbf{z} | \mathbf{x}, \mathbf{y}), p_\theta(\mathbf{z} | \mathbf{y})] + \log p_\theta(\mathbf{y})\end{aligned}$$

Joint VAE: Training objective

- The training objective is given by

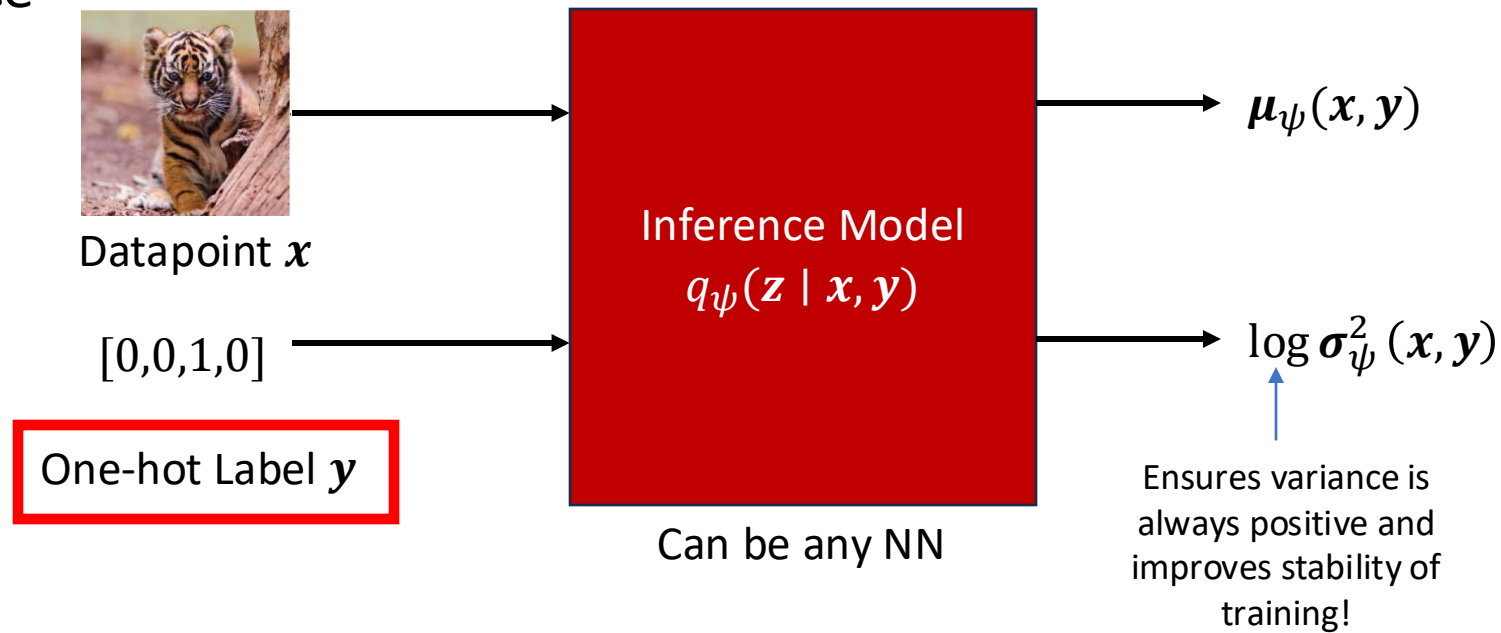
$$\begin{aligned} \max_{\theta} J_{\text{cond}} &:= \mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}} \log p_{\theta}(\mathbf{x}, \mathbf{y}) \\ &= \max_{\theta} \max_{\psi} \mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}} \left[\mathbb{E}_{q_{\psi}(\mathbf{z} | \mathbf{x}, \mathbf{y})} \log p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y}) - KL[q_{\psi}(\mathbf{z} | \mathbf{x}, \mathbf{y}) || p_{\theta}(\mathbf{z} | \mathbf{y})] + \log p_{\theta}(\mathbf{y}) \right] \end{aligned}$$

- What are $q_{\psi}(\mathbf{z} | \mathbf{x}, \mathbf{y})$, $p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y})$, $p_{\theta}(\mathbf{z} | \mathbf{y})$?
 - They are similar to what we needed in regular VAEs, with an additional input $\mathbf{y}$
- What is $p_{\theta}(\mathbf{y})$?
 - This is a new term denoting the prior probability on class labels
- Answer: next two slides

Joint VAE: Setup

- We have three models we need to define for VAE model

1. **Inference model $q_{\psi}(\mathbf{z} \mid \mathbf{x}, \mathbf{y})$:** We will define as $q_{\psi}(\mathbf{z} \mid \mathbf{x}, \mathbf{y}) = \mathcal{N}(\mathbf{z}; \mu_{\psi}(\mathbf{x}, \mathbf{y}), \text{diag}(\sigma_{\psi}^2(\mathbf{x})))$, i.e., a normal distribution with learned mean and covariance



2. **Latent prior:** $p(\mathbf{z} \mid \mathbf{y}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$

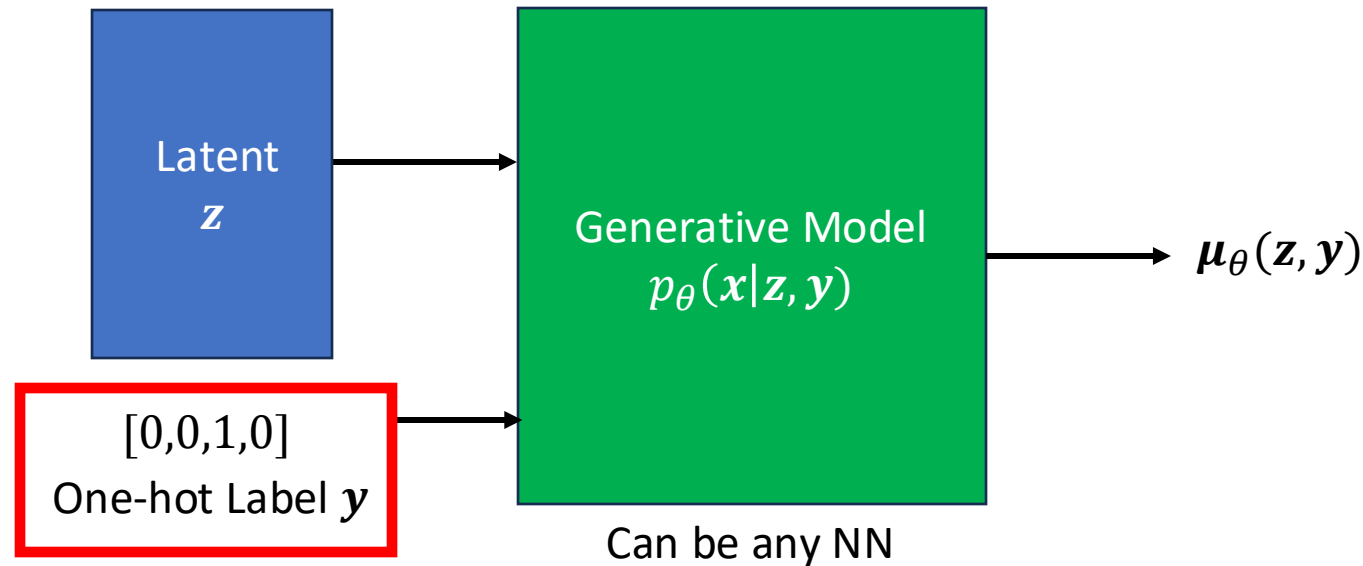
3. **Class prior:** $p_{\theta}(\mathbf{y}) = \text{Cat}(\pi)$

Joint VAE: Setup

- We have three models we need to define for VAE model

4. **Generative model $p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y})$** : We will define as

- $p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y}) = \mathcal{N}(\mathbf{x}; \mu_{\theta}(\mathbf{z}, \mathbf{y}), \eta^2 \mathbf{I})$, i.e., a normal distribution with learned mean and variance
- $p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y}) = \text{Cat}(\mathbf{x}; \pi_{\theta}(\mathbf{z}, \mathbf{y}))$, i.e., a categorical distribution with learned class probabilities

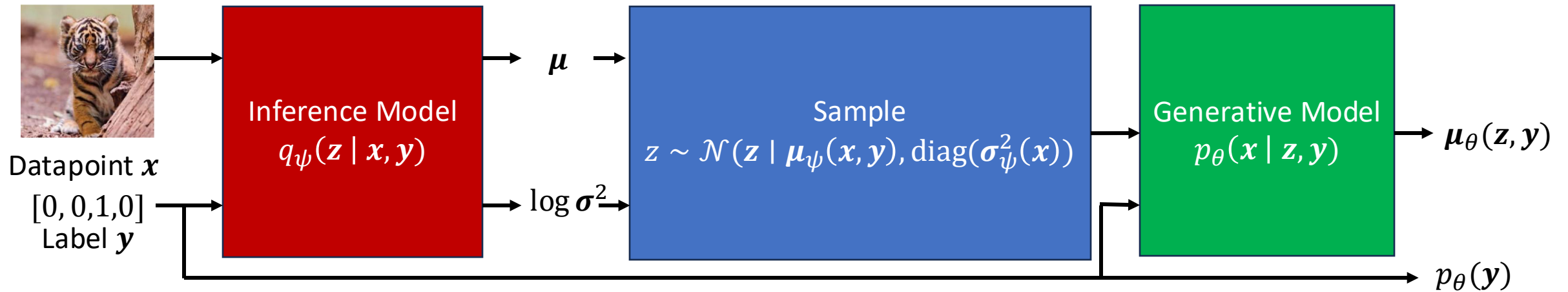


- Note this can be defined in many different ways, yielding different models (such as a categorical distribution over 255 values of each pixel)

Joint VAE: Training objective

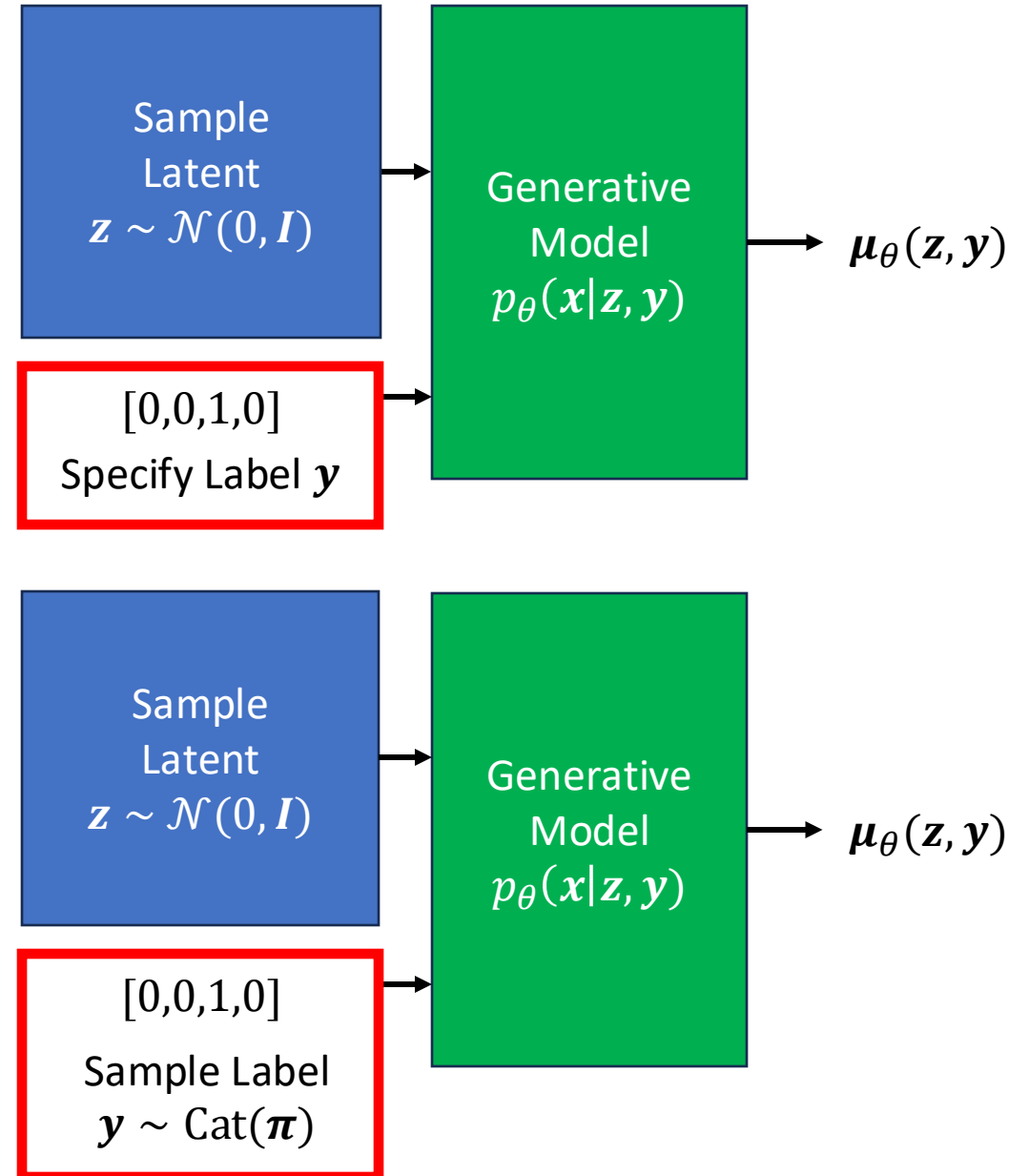
- So far, the training objective is given by

$$\begin{aligned} \max_{\theta} J_{\text{cond}} &:= \mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}} \log p_{\theta}(\mathbf{x}, \mathbf{y}) \\ &= \max_{\theta} \max_{\psi} \mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}} \left[\mathbb{E}_{q_{\psi}(\mathbf{z} | \mathbf{x}, \mathbf{y})} \log p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y}) - KL[q_{\psi}(\mathbf{z} | \mathbf{x}, \mathbf{y}) || p_{\theta}(\mathbf{z} | \mathbf{y})] + \log p_{\theta}(\mathbf{y}) \right] \end{aligned}$$



Joint VAE at inference

- At inference time, we can do either:
 - Conditional Generation:
 - Specify a label $\mathbf{y}$
 - Sample from $p(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$
 - Sample from $p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y})$
 - Joint Generation:
 - Sample a label $\mathbf{y} \sim p_{\theta}(\mathbf{y}) = \text{Cat}(\boldsymbol{\pi})$
 - Sample from $p(\mathbf{z})$, which is a Gaussian $\mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$
 - Sample from $p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y})$



Joint VAE: Training objective

- So far, the training objective is given by

$$\begin{aligned} & \max_{\theta} \mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}} \log p_{\theta}(\mathbf{x}, \mathbf{y}) \\ &= \max_{\theta} \max_{\psi} \mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}} \left[\mathbb{E}_{q_{\psi}(\mathbf{z} | \mathbf{x}, \mathbf{y})} \log p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y}) - KL[q_{\psi}(\mathbf{z} | \mathbf{x}, \mathbf{y}) || p_{\theta}(\mathbf{z} | \mathbf{y})] + \log p_{\theta}(\mathbf{y}) \right] =: J_{\text{cond}} \end{aligned}$$

- But we cannot calculate the distribution of $\mathbf{y}$ given $\mathbf{x}$ (classification)
- How can we also model classification?
 - Again, let us start with MLE principle: $\mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}} \log q_{\psi}(\mathbf{x}, \mathbf{y})$
 - Define $q_{\psi}(\mathbf{x}, \mathbf{y}) = q_{\psi}(\mathbf{y} | \mathbf{x}) p_{\text{data}}(\mathbf{x})$, where $q_{\psi}(\mathbf{y} | \mathbf{x})$ is a classifier
 - $\mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}} \log q_{\psi}(\mathbf{x}, \mathbf{y}) = \mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}} \log q_{\psi}(\mathbf{y} | \mathbf{x}) + \text{const. not depending on } \psi$
 - This gives the classification loss $J_{\text{cls}} := \mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}} \log q_{\psi}(\mathbf{y} | \mathbf{x})$
 - Requires introducing new parameters $q_{\psi}(\mathbf{y} | \mathbf{x})$
- The final objective is given by $J_{\text{cond}} + \lambda J_{\text{cls}}$
 - They originate from the MLE principles, since it is the sum of two log-likelihoods

Joint VAE: Summary

$$\max_{\theta} \max_{\psi} \mathbb{E}_{\mathbf{x}, \mathbf{y} \sim p_{\text{data}}} \left[\mathbb{E}_{q_{\psi}(\mathbf{z} | \mathbf{x}, \mathbf{y})} \log p_{\theta}(\mathbf{x} | \mathbf{z}, \mathbf{y}) - KL[q_{\psi}(\mathbf{z} | \mathbf{x}, \mathbf{y}) || p_{\theta}(\mathbf{z} | \mathbf{y})] + \log p_{\theta}(\mathbf{y}) + \lambda \log q_{\psi}(\mathbf{y} | \mathbf{x}) \right]$$

